

Zhiou Zhang

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Summary

Results-driven IT professional with hands-on experience delivering AI and data-driven systems. Skilled in Python, machine learning pipelines, computer vision, natural language processing, and RAG frameworks. Passionate about solving complex problems with practical AI solutions, and focused on automating workflows and designing maintainable, scalable architectures that supports business needs.

Skills

Languages:	Python, Rust
ML/DL:	pandas, numpy, scikit-learn, PyTorch, TensorFlow, LSTM
LLM Applications:	RAG (LangChain, Gemini, Chroma), prompt engineering
Database:	SQL, PostgreSQL
Data Visualization:	Matplotlib, OpenCV
Tools:	Git, the command line, Docker, Jira

Projects

AI Chatbot for UNSW Open Day

June 2025 – Now

Full-Stack Developer

Description: Designed and deployed an LLM-powered chatbot system for UNSW CSE Open Day, using Retrieval-Augmented Generation (RAG) to answer queries about academic programs with real-time source citation.

- Built an end-to-end **RAG pipeline** using **LangChain**, **Gemini 2.5**, and **ChromaDB** to enable semantic and keyword-based hybrid search over scraped and PDF content.
- Developed a **Flask-based RESTful API** with secure admin endpoints, feedback collection, and vector store updates via JWT authentication.
- Created a **Vue.js 3** frontend with session-aware chat interface, dynamic feedback collection, and admin dashboard for file/query management.
- Designed a lightweight **difflib-based cache** to reduce redundant Gemini calls, improving latency and API cost-efficiency.
- Built a scalable scraping system with **Selenium + BeautifulSoup**, supporting auto-discovery, change monitoring, and JSONL knowledge base updates.
- Containerized the full stack with **Docker** and deployed behind **nginx**, using separate configs for development and production environments.

House Price & Property Type Prediction

Mar 2025 – Apr 2025

Data Scientist & Machine Learning Engineer

Description: Built end-to-end ML pipelines to predict house prices and classify property types from heterogeneous property data.

- Built preprocessing workflows to handle missing values, multi-level median imputations, and expanded categorical features using **pandas**
- Trained a **GradientBoostingRegressor** on log-transformed prices, selecting top correlated features to reduce MAE and improve prediction stability
- Implemented a **RandomForestClassifier** with mutual information-based feature selection, optimizing weighted F1-score for imbalanced classes
- Automated experiment pipeline to generate submission-ready CSV outputs for both regression and classification

tasks

Traffic Volume Forecasting with Hybrid LSTM
Deep Learning Engineer

Jan 2025 - Feb 2025

Description: Developed a deep learning model to predict hourly traffic volumes based on multi-day and intra-day patterns, supporting better planning of RSU locations.

- Designed a dual-input **LSTM** model that combined day-level and intra-day traffic data to improve short-term forecasts
- Prepared rolling window datasets with MinMax scaling and handled cyclic time sequences to capture recurring patterns
- **Fine-tuned** dropout, L2 regularization and learning rates to reduce RMSE on validation sets, with early stopping to avoid overfitting
- Visualized prediction results, smoothed error curves and loss trends to check model reliability
- Provided traffic trend insights to inform more effective placement of roadside units (RSUs)

Food Semantic Segmentation for Calorie Estimation
Deep Learning Engineer

Oct 2024 – Nov 2024

Description: Developed a segmentation pipeline to identify food regions in images using YOLOv8 and DeepLabV3+, supporting portion size estimates for dietary analysis.

- Trained **DeepLabV3+** and **YOLOv8** models on the UEC FoodPix dataset, measuring standard and weighted mIoU across 102 food classes
- **Fine-tuned** hyperparameters including learning rates, augmentations and regularization to enhance segmentation accuracy
- Developed automated visualization tools with **OpenCV** and **Matplotlib** to overlay predictions, simplifying model comparison

Education

University of New South Wales, Sydney, Australia

Sept 2023 - Aug 2025 (expected)

Master of Information Technology – Data Science & Engineering

WAM: 76

Relevant Coursework: Data Structures & Algorithms, Database Systems, Software Construction, Neural Networks & Deep Learning, Computer Vision, Data Services Engineering, Database Systems Implementation, Statistical Inference, Information Retrieval & Web Search, Computer Networks & Applications, Web Front-End Programming, Project Management